

Problema #4 Conductores en paralelo con ρ distintas

$$(a) \quad V = IR, \quad R = R_1 + R_2.$$

$$R_1 = \frac{V}{I} = \frac{1}{I} \int_0^L \vec{E} d\vec{l}, \quad E = \rho \vec{J} = \rho \frac{I}{S} \hat{l}$$

$$R_1 = \frac{1}{I} \int_0^L \rho \frac{I}{S} dl$$

$$R_1 = \int_0^L \rho \frac{1}{S} dl = \left[\rho_1 \frac{L}{S} \right]$$

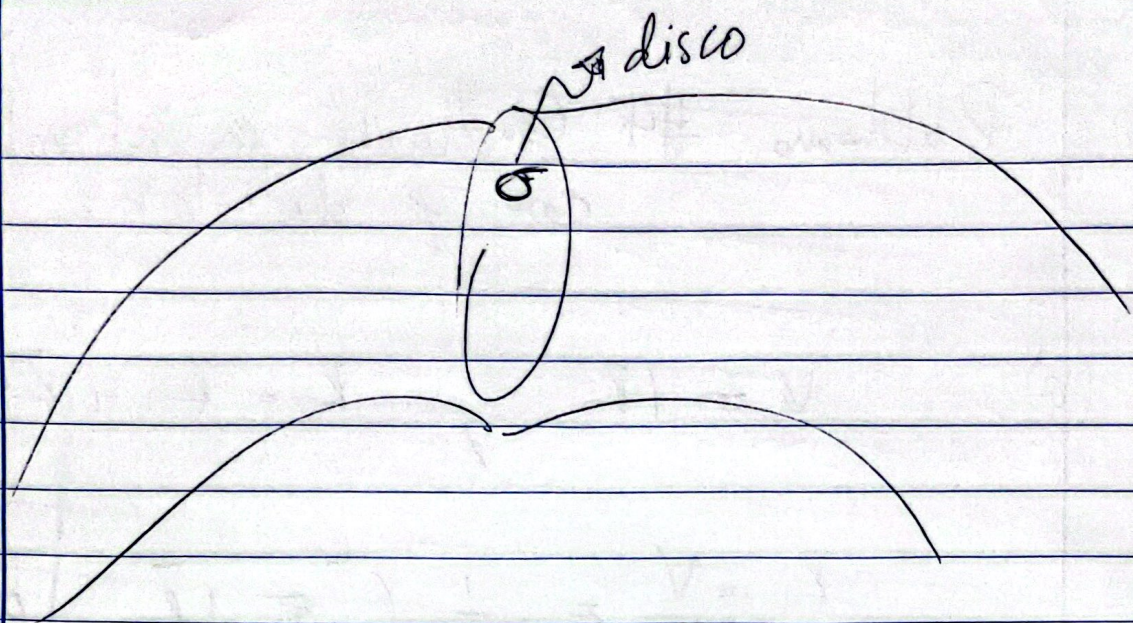
Similamente

$$\boxed{R_2 = \rho_2 \frac{L}{S}}$$

Entonces

$$I = \frac{V_{AB}}{R} = \frac{V_{AB}}{\frac{L}{S} (\rho_1 + \rho_2)} = \left[\frac{S V_{AB}}{L (\rho_1 + \rho_2)} \right]$$

(b)



El campo en el disco es

$$E_2 - E_1 = \frac{\sigma}{\epsilon_0} \quad (1)$$

$$E = \rho J, \quad J = \frac{I}{S}$$

$$\Rightarrow E = \rho \frac{I}{S} \quad (2)$$

② en ①

$$\rho_2 \frac{I}{S} - \rho_1 \frac{I}{S} = \frac{\sigma}{\epsilon_0}$$

$$\Rightarrow \sigma = \frac{\epsilon_0 I}{S} (\rho_2 - \rho_1)$$

$$\frac{Q}{S} = \sigma \rightarrow Q = \sigma S$$

$$\rightarrow Q = \epsilon_0 I (\rho_2 - \rho_1)$$